Omar Elgeoushy

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EDUCATION

Master of Science (MS) in Electrical and Computer Engineering August 2023 - May 2025
Purdue University, Indianapolis, IN

Bachelor of Science (BS) in Computer Science Graduated December 2022
Virginia Tech, Blacksburg, VA
GPA: 3.48/4.00 | Mathematics Minor | Dean's Honor List: Spring 2021, Fall 2021 and Spring 2022

SKILLS

Programming Skills: Java, Python, C, C++, C#, Ruby, React, JavaScript, Angular, MATLAB, CSS, Spring, SQL, PyTorch, .NET, Swift
Networking Skills: Broadband Network, TCP/IP, PON, SDN, VPN, VLANs, SNMP
Other Skill: Github, Jira, GDB, Valgrind, Linux, Unix, AWS, Data-structure, Scrum, AutoGPT, Agile, Waterfall, AI, Docker
Language Skills: Fluent in English, German, and Arabic

WORK EXPERIENCE

Software Engineering Research Assistant at Purdue, Indianapolis, IN August 2023 – Present
Autonomous Off-Road Ground Vehicle Simulator:

- Developing a Python-based AI model by utilizing a dataset of over 2500 images with TensorFlow and DeepLab to facilitate object detection and semantic segmentation within the autonomy stack, a pivotal element for self-driving cars
- Integrating the model seamlessly into the Multibody Dynamics & Terramechanics (IVRESS) system, thereby advancing autonomous vehicle technology
- Utilizing Unreal Engine 5 to visualize testing, enhancing the practical application of the technology

Software Engineer Intern at Si-Ware Systems, Menlo Park, CA June 2023 – September 2023

- Designed and developed a Python-based chatbot using AutoGPT API
- Developed automated assistance and enhanced communication to deliver a seamless user experience on the front-end with React integration based on custom data fed to the model
- Utilized Pinecone to implement a vector database, improving search and retrieval processes for the chatbot

Software Engineer Intern at 5APlus NetArch, Cheyenne, WY June – August 2021 and 2022
Network Monitoring Dashboard: June 2022 – August 2022

- Created a Python-based network monitoring dashboard using SNMP and TCP/IP to analyze data from network devices
- Implemented Angular and CSS for a user-friendly network performance monitoring interface
- Stored network data in a MySQL database for easy retrieval and analysis

Finance App: June 2021 – August 2021

- Contributed to the development of the Finance app, enabling payment processing using cryptocurrency for users
- Collaborated with the development team to implement and test features for the app
- Focused on backend development using Springboot and utilized SQL to manage and manipulate the database

Undergrad Teacher Assistant (UTA) | Computer Science Department at Virginia Tech August 2022 – December 2022

- Assisted 300+ students in achieving success in CS course and providing tutoring to students who needed extra support

PROJECTS

Case Study Library:

- Designed and developed a website hosted on AWS to serve as a resource for CS students at Virginia Tech, utilizing various AWS tools including Amplify, S3, AppSync, OpenSearch, and DynamoDB
- Implemented an upload page that allows students to upload their own presentations to the website
- Senior project website featuring over 850 presentations for viewing, accessible at https://casestudies.cs.vt.edu/